CHAKRADHAR SWAMIREDDY

GENERATIVE AI / RAG ENGINEER | PYTHON | LANGCHAIN | LLM APPLICATIONS

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PROFESSIONAL SUMMARY

Software Developer with 2+ years of production software development experience and hands-on project experience in Generative AI and RAG. Built a PDF question-answering system using LangChain, Pinecone and Groq LLaMA, with document chunking, metadata indexing, embeddings, hybrid BM25 + vector retrieval, query rewriting and ReAct-based question answering. Backend foundation in Python, FastAPI, REST APIs, SQL, Node.js and asynchronous processing.

TECHNICAL SKILLS

Programming: Python, JavaScript (ES6+), TypeScript, SQL

Generative AI / RAG: LLMs, Generative AI, prompt engineering, LangChain, RAG pipelines, document chunking, embeddings, semantic search, hybrid retrieval (BM25 + vector), query rewriting, ReAct agents, grounding verification

LLM / Vector: Groq LLaMA, OpenAI-compatible APIs, Pinecone, vector indexing, metadata filtering

Backend / Data: FastAPI, REST APIs, Node.js, Express.js, NestJS, PostgreSQL, MongoDB, MySQL, Redis, BullMQ

Tools: Git, VS Code, Playwright, Jest, Sentry, CI/CD, Vercel

AI PROJECTS

AI RAG Document Intelligence System | LangChain, Pinecone, Groq LLaMA, Next.js

• Built a PDF Q&A application with section-aware chunking, metadata indexing and embedding-based retrieval to provide context-grounded LLM responses.

• Combined vector similarity and BM25 keyword search using 0.6 / 0.4 weighting; added query rewriting, multi-hop decomposition, ReAct agent mode and grounding verification.

• Implemented LRU caching, streaming responses and trace-based observability. GitHub | Live demo

Web Query AI | Node.js, BullMQ, Redis, Playwright

• Built an AI-powered web querying backend with concurrent scraping jobs, dynamic content extraction and real-time progress updates; supports 20+ simultaneous background jobs.

GitHub: WebQuery-AI-Backend

PROFESSIONAL EXPERIENCE

Software Developer | EduNXT | Mar 2026 – Present

• Develop enterprise features and backend workflows using TypeScript, Node.js, REST APIs, SQL and database-driven services.

• Built asynchronous import / export workflows using queues and workers; implemented RBAC and multi-tenant application patterns and resolved authentication, API and database issues.

Frontend Developer | Accountiez Business Solutions | Dec 2024 – Dec 2025

• Built production applications using React, TypeScript, Node.js, REST APIs and MongoDB. Developed DesignCliq catalog workflows, reducing manual cataloging effort by 35%.

• Engineered DesignVault, a multi-client B2B catalog with reusable components and RBAC; API and data-flow optimization improved performance by 25%.

• Delivered CRM applications and a social media analytics dashboard with modular forms, monitoring and performance improvements.

Software Developer Intern | IIT Bombay (ASC) | Dec 2023 – Dec 2024

• Built a React / Chart.js dashboard with dynamic filters; improved Java/JSP security and helped consolidate 4+ React applications into a micro-frontend shell, reducing initial load time by 30%.

EDUCATION

B.Tech, Computer Science and Engineering | RGUKT, IIIT Srikakulam | 2019–2023 | CGPA: 8.6 / 10